

4 The structures of blood vessels relate to their functions.

The table gives information about some blood vessels in a dog.

Blood vessel	Number in body	Diameter / cm	Length of vessel / cm	Total surface area / cm ²	Velocity of blood flow / cm s ⁻¹
aorta	1	1.0	40.0	1.3×10^2	28.0
large arteries	40	0.3	20.0	7.5×10^2	7.8
capillaries	1.2×10^9	8.0×10^{-4}	0.1	3.0×10^5	3.6×10^{-2}
large veins	40	0.6	20.0		1.9
vena cava	1	1.25		1.6×10^2	18.0

(a) Complete the table.

Use the formula:

$$\text{surface area} = 2\pi rl$$

where r is the radius of the blood vessel and l is the length of the vessel.

(3)



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(b) Explain why the velocity of blood flow in the large arteries is slower than the velocity of blood flow in the aorta.

(3)

(c) Explain why the total surface area of the capillaries needs to be so high.

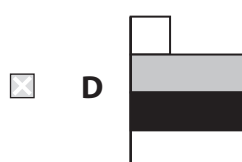
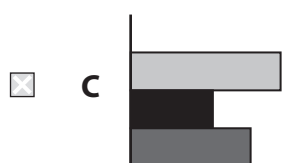
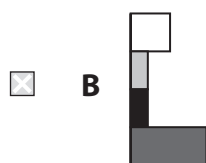
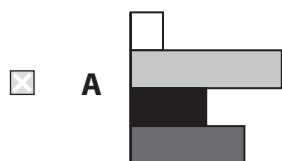
(2)



- (d) (i) The diagrams show the relative proportions of components found in blood vessel walls.

Which diagram represents the wall of a large artery?

(1)

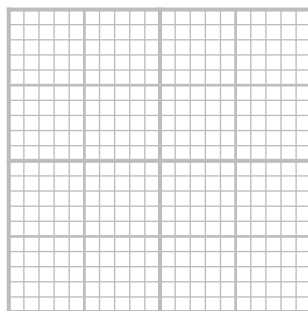


Key:

- ☐ endothelial cell layer
- ☐ elastic fibres
- ☐ smooth muscle
- ☐ collagen fibres

- (ii) Draw a diagram, using the style above, to represent the wall of a capillary.

(1)



(Total for Question 4 = 10 marks)

